

Modelos Generativos: Difusão

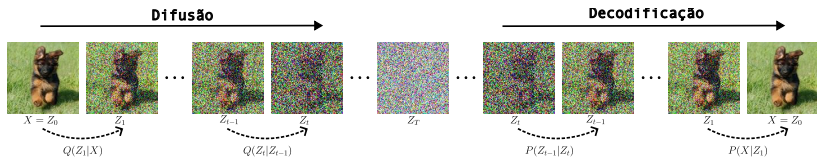
Machine Learning 2 2026.1

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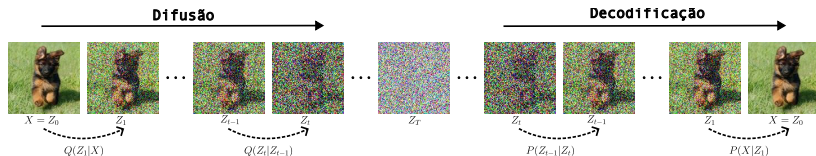
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Difusão: Overview



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- ▶ $Q(Z_t)$ modela o processo de difusão.
- ▶ $P(Z_t | \theta)$ modela o processo reverso de difusão.
- ▶ $X \sim Q(X)$ é a distribuição de dados reais.
- ▶ $Z_T \sim Q(Z_T) = \mathcal{N}(0, I)$ é a distribuição de ruído.

Definições: Encoder (Difusor)

$$X = Z_0 \rightarrow Z_1 \dots \rightarrow Z_t \rightarrow \dots \rightarrow Z_T$$

Assumindo a difusão como o resultado de uma gaussiana temos...

$$\begin{array}{ll} Z_1 \sim \mathcal{N}(\sqrt{1 - \beta_1} Z_0, \beta_1 I) & Z_1 = \sqrt{1 - \beta_1} Z_0 + \sqrt{\beta_1} \epsilon_1 \\ \vdots & \vdots \\ Z_t \sim \mathcal{N}(\sqrt{1 - \beta_t} Z_{t-1}, \beta_t I) & Z_t = \sqrt{1 - \beta_t} Z_{t-1} + \sqrt{\beta_t} \epsilon_t \\ \vdots & \vdots \\ Z_T \sim \mathcal{N}(\sqrt{1 - \beta_T} Z_{T-1}, \beta_T I) & Z_T = \sqrt{1 - \beta_T} Z_{T-1} + \sqrt{\beta_T} \epsilon_T \end{array}$$

Onde $\epsilon_t \sim \mathcal{N}(0, I)$ e $0 < \beta_t < 1$, para $t = 1, \dots, T$, são hiperparâmetros (*noise schedule*).

Convergência I

Afirmção: Para T suficientemente grande, $Z_T \sim \mathcal{N}(0, I)$.

$$\begin{aligned}\mathbb{E}[Z_t] &= \mathbb{E}[\sqrt{1 - \beta_t} Z_{t-1} + \sqrt{\beta_t} \epsilon_t] \\&= \sqrt{1 - \beta_t} \mathbb{E}[Z_{t-1}] + \sqrt{\beta_t} \mathbb{E}[\epsilon_t] \\&= \sqrt{1 - \beta_t} \mathbb{E}[\sqrt{1 - \beta_{t-1}} Z_{t-2} + \sqrt{\beta_{t-1}} \epsilon_{t-1}] + \sqrt{\beta_t} \mathbb{E}[\epsilon_t] \\&= \sqrt{1 - \beta_t} \sqrt{1 - \beta_{t-1}} \mathbb{E}[Z_{t-2}] + \underbrace{\sqrt{1 - \beta_t} \sqrt{\beta_{t-1}} \mathbb{E}[\epsilon_{t-1}] + \sqrt{\beta_t} \mathbb{E}[\epsilon_t]}_{=0, \text{ pois } \mathbb{E}[\epsilon_k]=0}\end{aligned}$$

...

$$= \prod_{k=0}^t \sqrt{1 - \beta_k} \mathbb{E}[Z_0]$$

E como $\lim_{T \rightarrow \infty} \prod_{k=0}^T \sqrt{1 - \beta_k} = 0$, temos que $\mathbb{E}[Z_t] \rightarrow 0$ quando $T \rightarrow \infty$.

Convergência I

$$\begin{aligned}\text{Var}[Z_t] &= \text{Var}[\sqrt{1 - \beta_t} Z_{t-1} + \sqrt{\beta_t} \epsilon_t] \\ &= (1 - \beta_t) \text{Var}[Z_{t-1}] + 2\sqrt{(1 - \beta_t)\beta_t} \text{Cov}[Z_{t-1}, \epsilon_t] + \beta_t \text{Var}[\epsilon_t] \\ &= (1 - \beta_t) \text{Var}[Z_{t-1}] + \beta_t \text{Var}[\epsilon_t] \\ &= (1 - \beta_t) \text{Var}[Z_{t-1}] + \beta_t I\end{aligned}$$

Pois, Z_{t-1} e ϵ_t são independentes e portanto $\text{Cov}[Z_{t-1}, \epsilon_t] = 0$.

Prova da independência: $Z_{t-1} = f(Z_0, \epsilon_1, \dots, \epsilon_{t-1})$ depende apenas de $\epsilon_1, \dots, \epsilon_{t-1}$, que são independentes de ϵ_t por construção. Logo $Z_{t-1} \perp \epsilon_t$.

Convergência II

$$\begin{aligned}\text{Var}[Z_t] &= (1 - \beta_t) ((1 - \beta_{t-1}) \text{Var}[Z_{t-2}] + \beta_{t-1}I) + \beta_t I \\&= (1 - \beta_t)(1 - \beta_{t-1}) \text{Var}[Z_{t-2}] + (1 - \beta_t)\beta_{t-1}I + \beta_t I \\&= (1 - \beta_t)(1 - \beta_{t-1}) \text{Var}[Z_{t-2}] + (1 - \beta_t)\beta_{t-1}I + \beta_t I + I - I \\&= (1 - \beta_t)(1 - \beta_{t-1}) \text{Var}[Z_{t-2}] - (1 - \beta_t)(1 - \beta_{t-1})I + I \\&\vdots \\&= \prod_{k=0}^t (1 - \beta_k) \text{Var}[Z_0] - \prod_{k=0}^t (1 - \beta_k)I + I\end{aligned}$$

portanto,

$$\begin{aligned}\lim_{T \rightarrow \infty} \text{Var}[Z_t] &= \lim_{T \rightarrow \infty} \left(\prod_{k=0}^T (1 - \beta_k) \text{Var}[Z_0] - \prod_{k=0}^T (1 - \beta_k)I + I \right) \\&= 0 - 0 + I = I\end{aligned}$$

Podemos escrever:

$$Z_1 \sim Q(Z_1|X) = \mathcal{N}(\sqrt{1 - \beta_1}Z_0, \beta_1 I)$$

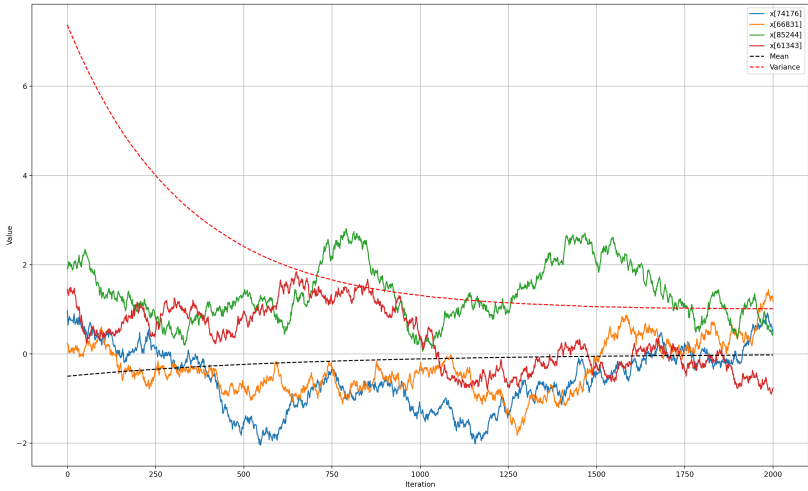
$$\vdots$$

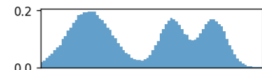
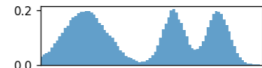
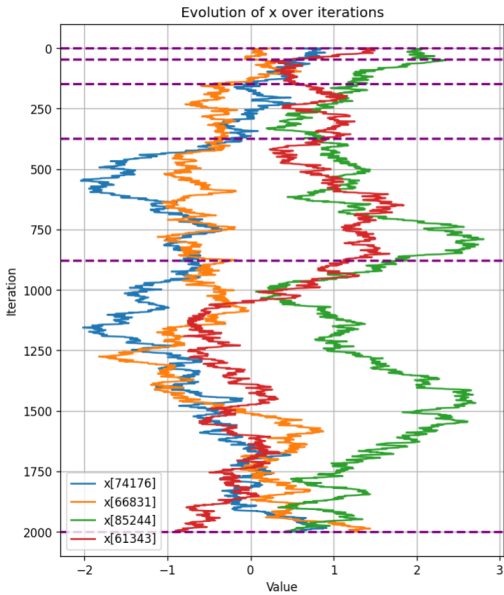
$$Z_t \sim Q(Z_t|Z_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t}Z_{t-1}, \beta_t I)$$

$$\vdots$$

$$Z_T \sim Q(Z_T|Z_{T-1}) = \mathcal{N}(\sqrt{1 - \beta_T}Z_{T-1}, \beta_T I) \approx \mathcal{N}(0, I)$$

Evolution of x over iterations





Kernel de Difusão I

Fixado $Z_0 = X$, a distribuição de Z_t pode ser encontrada de forma analítica:

$$\begin{aligned}Z_1 &= \sqrt{1 - \beta_1} Z_0 + \sqrt{\beta_1} \epsilon_1 \\Z_2 &= \sqrt{1 - \beta_2} Z_1 + \sqrt{\beta_2} \epsilon_2 \\&= \sqrt{1 - \beta_2} \left(\sqrt{1 - \beta_1} Z_0 + \sqrt{\beta_1} \epsilon_1 \right) + \sqrt{\beta_2} \epsilon_2 \\&= \sqrt{(1 - \beta_2)(1 - \beta_1)} Z_0 + \sqrt{(1 - \beta_2)\beta_1} \epsilon_1 + \sqrt{\beta_2} \epsilon_2\end{aligned}$$

Como ϵ_1 e ϵ_2 são normais de média zero, temos que $\sqrt{(1 - \beta_2)\beta_1} \epsilon_1 + \sqrt{\beta_2} \epsilon_2$ é uma normal de média zero, cuja variância é dada por

$$\begin{aligned}\text{Var} \left[\sqrt{(1 - \beta_2)\beta_1} \epsilon_1 + \sqrt{\beta_2} \epsilon_2 \right] &= (1 - \beta_2)\beta_1 \text{Var}[\epsilon_1] + \beta_2 \text{Var}[\epsilon_2] \\&= (1 - \beta_2)\beta_1 I + \beta_2 I \\&= ((1 - \beta_2)\beta_1 + \beta_2) I \\&= (1 - (1 - \beta_2)(1 - \beta_1)) I\end{aligned}$$

Kernel de Difusão II

portanto

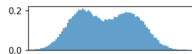
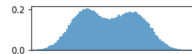
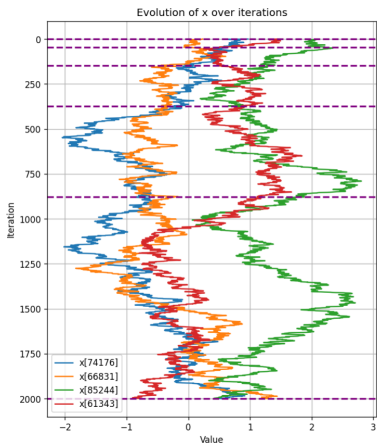
$$Z_2 = \sqrt{(1 - \beta_2)(1 - \beta_1)}Z_0 + \sqrt{(1 - (1 - \beta_2)(1 - \beta_1))}\epsilon, \text{ onde } \epsilon \sim \mathcal{N}(0, I)$$

ou seja,

$$Z_2 \sim Q(Z_2|Z_0) = \mathcal{N}\left(\sqrt{(1 - \beta_2)(1 - \beta_1)}Z_0, (1 - (1 - \beta_2)(1 - \beta_1))I\right)$$

De forma análoga, podemos mostrar que

$$\begin{aligned} Z_t \sim Q(Z_t|Z_0) &= \mathcal{N}\left(\sqrt{\prod_{k=1}^t (1 - \beta_k)}Z_0, \left(1 - \prod_{k=1}^t (1 - \beta_k)\right)I\right) \\ &= \mathcal{N}(\sqrt{\alpha_t}Z_0, 1 - \alpha_t I), \text{ onde } \alpha_t = \prod_{k=1}^t (1 - \beta_k) \end{aligned}$$



$$Q(Z_t)$$

$$\int Q(Z_t|X)Q(X)dX$$

Decoder (Difusor Reverso)

Queremos percorrer o caminho inverso, ou seja, dado Z_T encontrar X . Para isso, precisamos caracterizar a distribuição de Z_{t-1} dado Z_t .

$$Q(Z_{t-1}|Z_t) = \frac{Q(Z_t|Z_{t-1})Q(Z_{t-1})}{Q(Z_t)}$$

$Q(Z_t|Z_{t-1})$ é conhecida. Porém, não conhecemos $Q(Z_{t-1})$ e $Q(Z_t)$, já que

$$Q(Z_t) = \int Q(Z_t|X)Q(X)dX,$$

e $Q(X)$, que é a distribuição real dos dados, é desconhecida.

Porém se conhecemos X e Z_t , podemos escrever:

$$Q(Z_{t-1}|Z_t, X) = \frac{Q(Z_t|Z_{t-1}, X) Q(Z_{t-1}|X)}{Q(Z_t|X)} \quad (18.12)$$

$$\propto Q(Z_t|Z_{t-1}) Q(Z_{t-1}|X)$$

(i) $Q(Z_t|Z_{t-1}, X) = Q(Z_t|Z_{t-1})$ (Markov) (ii) $Q(Z_t|X)$ é constante em Z_{t-1}

$$= \mathcal{N}_{Z_t} \left[\sqrt{1 - \beta_t} \cdot Z_{t-1}, \beta_t I \right] \mathcal{N}_{Z_{t-1}} \left[\sqrt{\alpha_{t-1}} \cdot X, (1 - \alpha_{t-1}) I \right]$$

$$\propto \mathcal{N}_{Z_{t-1}} \left[\frac{1}{\sqrt{1 - \beta_t}} Z_t, \frac{\beta_t}{1 - \beta_t} I \right] \mathcal{N}_{Z_{t-1}} \left[\sqrt{\alpha_{t-1}} \cdot X, (1 - \alpha_{t-1}) I \right]$$

$$\mathcal{N}_v[Aw, B] \propto \mathcal{N}_w \left[(A^T B^{-1} A)^{-1} A^T B^{-1} v, (A^T B^{-1} A)^{-1} \right]$$

$$\mathcal{N}_w[a, A] \cdot \mathcal{N}_w[b, B] \propto \mathcal{N}_w \left[(A^{-1} + B^{-1})^{-1} (A^{-1}a + B^{-1}b), (A^{-1} + B^{-1})^{-1} \right]$$

$$Q(Z_{t-1}|Z_t, X) = \mathcal{N}_{Z_{t-1}} \left[\left(\frac{1 - \alpha_{t-1}}{1 - \alpha_t} \right) \sqrt{1 - \beta_t} Z_t + \frac{\sqrt{\alpha_{t-1}} \beta_t}{1 - \alpha_t} X, \frac{\beta_t (1 - \alpha_{t-1})}{1 - \alpha_t} I \right]$$

Decoder (Difusor Reverso)

Aproximamos $Q(Z_{t-1}|Z_t, X)$ por $P(Z_{t-1}|Z_t, \theta)$, onde $\theta = (\theta_1, \dots, \theta_T)$ são os parâmetros do modelo.

$$P(Z_T) = \mathcal{N}(0, I)$$

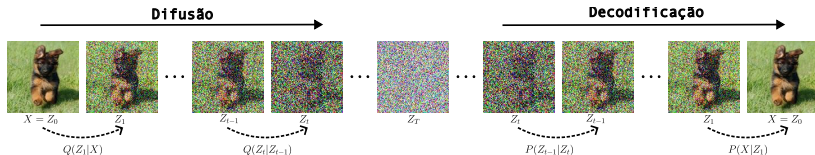
$$\vdots$$

$$P(Z_{t-1}|Z_t, \theta) = \mathcal{N}_{Z_{t-1}} [f_t(Z_t, \theta_t), \sigma_t^2 I]$$

$$\vdots$$

$$P(X|Z_1, \theta) = \mathcal{N}_X [f_1(Z_1, \theta_1), \sigma_1^2 I]$$

Decoder (Difusor Reverso)



Resumindo, temos que

- ▶ $Q(Z_t|X) = \mathcal{N}(\sqrt{\alpha_t}X, (1 - \alpha_t)I)$, onde $\alpha_t = \prod_{k=1}^t (1 - \beta_k)$.
- ▶ $Q(Z_{t-1}|Z_t, X) = \mathcal{N}_{Z_{t-1}} \left[\left(\frac{1 - \alpha_{t-1}}{1 - \alpha_t} \right) \sqrt{1 - \beta_t} Z_t + \frac{\sqrt{\alpha_{t-1}} \beta_t}{1 - \alpha_t} X, \frac{\beta_t (1 - \alpha_{t-1})}{1 - \alpha_t} I \right]$
- ▶ $P(Z_{t-1}|Z_t, \theta) = \mathcal{N}_{Z_{t-1}} [f_t(Z_t, \theta_t), \sigma_t^2 I]$

ELBO

$$\begin{aligned} D_{\text{KL}} [Q(Z_{1...T}|X) \parallel P(Z_{1...T}|X, \theta)] \\ &= \mathbb{E}_{Q(Z_{1...T}|X)} [\log Q(Z_{1...T}|X) - \log P(Z_{1...T}|X, \theta)] \\ &= \mathbb{E}_{Q(Z_{1...T}|X)} \left[\log Q(Z_{1...T}|X) - \log \frac{P(Z_{1...T}, X|\theta)}{P(X|\theta)} \right] \\ &= \mathbb{E}_{Q(Z_{1...T}|X)} [\log Q(Z_{1...T}|X) - \log P(Z_{1...T}, X|\theta)] + \log P(X|\theta) \end{aligned}$$

Portanto

$$\begin{aligned} \log P(X|\theta) \\ &= D_{\text{KL}} [Q(Z_{1...T}|X) \parallel P(Z_{1...T}|X, \theta)] + \mathbb{E}_{Q(Z_{1...T}|X)} \left[\log \frac{P(Z_{1...T}, X|\theta)}{Q(Z_{1...T}|X)} \right] \\ &= \underbrace{D_{\text{KL}} [Q(Z_{1...T}|X) \parallel P(Z_{1...T}|X, \theta)]}_{\geq 0} + \mathbb{E}_{Q(Z_{1...T}|X)} \left[\log \frac{P(Z_{1...T}, X|\theta)}{Q(Z_{1...T}|X)} \right] \\ &\geq \mathbb{E}_{Q(Z_{1...T}|X)} \left[\log \frac{P(Z_{1...T}, X|\theta)}{Q(Z_{1...T}|X)} \right] = \text{ELBO} [\theta] \end{aligned}$$

Simplificação da ELBO I

$$\begin{aligned}P(X, Z_{1\dots T}|\theta) &= \underbrace{P(X|Z_{1\dots T}, \theta)}_{\text{Markov}} P(Z_{1\dots T}|\theta) \\&= P(X|Z_1, \theta) P(Z_{1\dots T}|\theta) \\&= P(X|Z_1, \theta) P(Z_1|Z_{2\dots T}, \theta) P(Z_{2\dots T}|\theta) \\&= P(X|Z_1, \theta) P(Z_1|Z_2, \theta) P(Z_{2\dots T}|\theta) \\&\dots \\&= P(X|Z_1, \theta) \prod_{t=2}^T P(Z_{t-1}|Z_t, \theta) P(Z_T|\theta)\end{aligned}$$

Simplificação da ELBO II

$$\begin{aligned} Q(Z_{1\dots T}|X) &= Q(Z_1|X) \underbrace{Q(Z_{2\dots T}|Z_1, X)}_{\text{Markov}} \\ &= Q(Z_1|X)Q(Z_{2\dots T}|Z_1) \\ &= Q(Z_1|X)Q(Z_2|Z_1)Q(Z_{3\dots T}|Z_2) \\ &\dots \\ &= Q(Z_1|X) \prod_{t=2}^T Q(Z_t|Z_{t-1}) \end{aligned}$$

Simplificação da ELBO III

Resumindo, temos que

$$P(X, Z_{1...T}|\theta) = P(X|Z_1, \theta) \prod_{t=2}^T P(Z_{t-1}|Z_t, \theta) P(Z_T|\theta)$$

$$Q(Z_{1...T}|X) = Q(Z_1|X) \prod_{t=2}^T Q(Z_t|Z_{t-1})$$

Tomando o log da razão entre as duas expressões,

$$\begin{aligned} \log \frac{P(Z_{1...T}, X|\theta)}{Q(Z_{1...T}|X)} &= \log \frac{P(X|Z_1, \theta) \prod_{t=2}^T P(Z_{t-1}|Z_t, \theta) P(Z_T|\theta)}{Q(Z_1|X) \prod_{t=2}^T Q(Z_t|Z_{t-1})} \\ &= \frac{\log P(X|Z_1, \theta)}{Q(Z_1|X)} + \log \frac{\prod_{t=2}^T P(Z_{t-1}|Z_t, \theta)}{\prod_{t=2}^T Q(Z_t|Z_{t-1})} + \log P(Z_T|\theta) \end{aligned}$$

Simplificação da ELBO IV

Mas

$$Q(Z_t|Z_{t-1}) = Q(Z_t|Z_{t-1}, X) = \frac{Q(Z_{t-1}|Z_t, X)Q(Z_t|X)}{Q(Z_{t-1}|X)}$$

Então,

$$\begin{aligned} & \log \frac{P(Z_{1...T}, X|\theta)}{Q(Z_{1...T}|X)} \\ &= \log \frac{P(X|Z_1, \theta)}{Q(Z_1|X)} + \log \frac{\prod_{t=2}^T P(Z_{t-1}|Z_t, \theta)Q(Z_{t-1}|X)}{\prod_{t=2}^T Q(Z_{t-1}|Z_t, X)Q(Z_t|X)} + \log P(Z_T|\theta) \\ &= \log P(X|Z_1, \theta) + \log \frac{\prod_{t=2}^T P(Z_{t-1}|Z_t, \theta)}{\prod_{t=2}^T Q(Z_{t-1}|Z_t, X)} + \log \frac{P(Z_T|\theta)}{Q(Z_T|X)} \\ &= \log P(X|Z_1, \theta) + \sum_{t=2}^T \log \frac{P(Z_{t-1}|Z_t, \theta)}{Q(Z_{t-1}|Z_t, X)} + \log \frac{P(Z_T|\theta)}{Q(Z_T|X)} \end{aligned}$$

Simplificação da ELBO V

Substituindo na ELBO, temos que

$$\begin{aligned}\text{ELBO}[\theta] &= \mathbb{E}_{Q(Z_{1...T}|X)} \left[\log \frac{P(Z_{1...T}, X|\theta)}{Q(Z_{1...T}|X)} \right] \\&= \mathbb{E}_{Q(Z_{1...T}|X)} [\log P(X|Z_1, \theta)] \\&\quad + \sum_{t=2}^T \mathbb{E}_{Q(Z_{1...T}|X)} \left[\log \frac{P(Z_{t-1}|Z_t, \theta)}{Q(Z_{t-1}|Z_t, X)} \right] \\&\quad + \mathbb{E}_{Q(Z_{1...T}|X)} \left[\log \frac{P(Z_T|\theta)}{Q(Z_T|X)} \right]\end{aligned}$$

Simplificação da ELBO VI

$$\begin{aligned}\mathbb{E}_{Q(Z_1 \dots T|X)} [\log P(X|Z_1, \theta)] &= \iint Q(Z_1 \dots T|X) \log P(X|Z_1, \theta) dZ_1 \dots T \\ &= \int \log P(X|Z_1, \theta) \left(\iint Q(Z_1 \dots T|X) dZ_2 \dots T \right) dZ_1 \\ &= \int \log P(X|Z_1, \theta) Q(Z_1|X) dZ_1 \\ &= \mathbb{E}_{Q(Z_1|X)} [\log P(X|Z_1, \theta)]\end{aligned}$$

Simplificação da ELBO VII

$$\begin{aligned}& \sum_{t=2}^T \mathbb{E}_{Q(Z_{1...T}|X)} \left[\log \frac{P(Z_{t-1}|Z_t, \theta)}{Q(Z_{t-1}|Z_t, X)} \right] \\&= \sum_{t=2}^T \iint Q(Z_{1...T}|X) \log \frac{P(Z_{t-1}|Z_t, \theta)}{Q(Z_{t-1}|Z_t, X)} dZ_{1...T} \\&= \sum_{t=2}^T \iint \log \frac{P(Z_{t-1}|Z_t, \theta)}{Q(Z_{t-1}|Z_t, X)} \left(\iint Q(Z_{1...T}|X) dZ_{\{\neq t-1, \neq t\}} \right) dZ_{t-1} dZ_t \\&= \sum_{t=2}^T \iint \log \frac{P(Z_{t-1}|Z_t, \theta)}{Q(Z_{t-1}|Z_t, X)} Q(Z_{t-1}, Z_t|X) dZ_{t-1} dZ_t \\&= \sum_{t=2}^T \iint \log \frac{P(Z_{t-1}|Z_t, \theta)}{Q(Z_{t-1}|Z_t, X)} Q(Z_{t-1}|Z_t, X) Q(Z_t|X) dZ_{t-1} dZ_t \\&= \sum_{t=2}^T \int \left(\int \log \frac{P(Z_{t-1}|Z_t, \theta)}{Q(Z_{t-1}|Z_t, X)} Q(Z_{t-1}|Z_t, X) dZ_{t-1} \right) Q(Z_t|X) dZ_t\end{aligned}$$

Simplificação da ELBO VIII

$$\begin{aligned} & \sum_{t=2}^T \mathbb{E}_{Q(Z_{1 \dots T} | X)} \left[\log \frac{P(Z_{t-1} | Z_t, \theta)}{Q(Z_{t-1} | Z_t, X)} \right] \\ &= \sum_{t=2}^T \int \left(\int \log \frac{P(Z_{t-1} | Z_t, \theta)}{Q(Z_{t-1} | Z_t, X)} Q(Z_{t-1} | Z_t, X) dZ_{t-1} \right) Q(Z_t | X) dZ_t \\ &= \sum_{t=2}^T \int D_{\text{KL}} [Q(Z_{t-1} | Z_t, X) \parallel P(Z_{t-1} | Z_t, \theta)] Q(Z_t | X) dZ_t \\ &= \sum_{t=2}^T \mathbb{E}_{Q(Z_t | X)} [D_{\text{KL}} [Q(Z_{t-1} | Z_t, X) \parallel P(Z_{t-1} | Z_t, \theta)]] \end{aligned}$$

Simplificação da ELBO IX

$$\begin{aligned}\mathbb{E}_{Q(Z_{1 \dots T}|X)} \left[\log \frac{P(Z_T|\theta)}{Q(Z_T|X)} \right] &= \mathbb{E}_{Q(Z_T|X)} \left[\log \frac{P(Z_T|\theta)}{Q(Z_T|X)} \right] \\ &= D_{\text{KL}} [Q(Z_T|X) \parallel P(Z_T|\theta)] \\ &= D_{\text{KL}} [\mathcal{N}(0, I) \parallel \mathcal{N}(0, I)] \\ &= 0\end{aligned}$$

Resumindo a ELBO I

$$\begin{aligned}\text{ELBO} [\theta] = & \mathbb{E}_{Q(Z_1|X)} [\log P(X|Z_1, \theta)] \\ & + \sum_{t=2}^T \mathbb{E}_{Q(Z_t|X)} [D_{\text{KL}} [Q(Z_{t-1}|Z_t, X) \parallel P(Z_{t-1}|Z_t, \theta)]]\end{aligned}$$

Conhecemos cada termo da ELBO de forma explícita:

$$P(X|Z_1, \theta) = \mathcal{N}_x(f_1(z_1, \theta_1), \sigma_1^2 I)$$

$$P(Z_{t-1}|Z_t, \theta) = \mathcal{N}_{z_{t-1}}(f_t(Z_t, \theta_t), \sigma_t^2 I)$$

$$Q(Z_{t-1}|Z_t, X) = \mathcal{N}_{z_{t-1}}\left(\frac{1 - \alpha_{t-1}}{1 - \alpha_t} \sqrt{1 - \beta_t} Z_t + \frac{\sqrt{\alpha_{t-1}} \beta_t}{1 - \alpha_t} X, \frac{\beta_t (1 - \alpha_{t-1})}{\alpha_t} I\right)$$

Resumindo a ELBO II

Considerando que:

$$\log P(X|Z_1, \theta) = -\frac{1}{2\sigma_1^2} \|X - f_1(Z_1, \theta_1)\|^2 + C$$

e

$$\begin{aligned} D_{\text{KL}} [Q(Z_{t-1}|Z_t, X) \| P(Z_{t-1}|Z_t, \theta)] \\ = \frac{1}{2\sigma_t^2} \left\| \frac{1 - \alpha_{t-1}}{1 - \alpha_t} \sqrt{1 - \beta_t} Z_t + \frac{\sqrt{\alpha_{t-1}\beta_t}}{1 - \alpha_t} X - f_t(Z_t, \theta_t) \right\|^2 + C \end{aligned}$$

ELBO como Loss Function I

Podemos escrever a ELBO como uma função de perda, ou seja, queremos minimizar a ELBO:

$$\begin{aligned}\mathcal{L}(\theta) &= -\text{ELBO}[\theta] \\ &= -\mathbb{E}_{Q(Z_1|X)} [\log P(X|Z_1, \theta)] \\ &\quad - \sum_{t=2}^T \mathbb{E}_{Q(Z_t|X)} [D_{\text{KL}} [Q(Z_{t-1}|Z_t, X) \parallel P(Z_{t-1}|Z_t, \theta)]] \\ &= \mathbb{E}_{Q(Z_1|X)} \left[\frac{1}{2\sigma_1^2} \|X - f_1(Z_1, \theta_1)\|^2 \right] \\ &\quad + \sum_{t=2}^T \mathbb{E}_{Q(Z_t|X)} \left[\frac{1}{2\sigma_t^2} \left\| \frac{1 - \alpha_{t-1}}{1 - \alpha_t} \sqrt{1 - \beta_t} Z_t + \frac{\sqrt{\alpha_{t-1}} \beta_t}{1 - \alpha_t} X - f_t(Z_t, \theta_t) \right\|^2 \right]\end{aligned}$$

ELBO como Loss Function II

Podemos eliminar as integrações ($\mathbb{E}_{Q(Z_t|X)}$) usando amostragem de Monte Carlo, ou seja, podemos amostrar Z_t de $Q(Z_t|X)$ e calcular a média. Considerando que temos diversas imagens de treino, podemos escrever a função de perda como:

$$\mathcal{L}(\theta) = \sum_{i=1}^I \frac{1}{N} \sum_{j=1}^N \left(\frac{1}{2\sigma_1^2} \|X^{(i)} - f_1(Z_1^{(i,j)}, \theta_1)\|^2 + \sum_{t=2}^T \frac{1}{2\sigma_t^2} \left\| \frac{1 - \alpha_{t-1}}{1 - \alpha_t} \sqrt{1 - \beta_t} Z_t^{(i,j)} + \frac{\sqrt{\alpha_{t-1}} \beta_t}{1 - \alpha_t} X^{(i)} - f_t(Z_t^{(i,j)}, \theta_t) \right\|^2 \right)$$

(Normalmente usamos $N = 1$)

Que já nos possibilita treinar o modelo! Mas podemos fazer melhor!

Reparametrização da Loss Function I

Sabemos que

$$Z_t = \sqrt{\alpha_t}X + \sqrt{1 - \alpha_t}\epsilon,$$

portanto podemos escrever

$$X = \frac{1}{\sqrt{\alpha_t}}Z_t - \frac{\sqrt{1 - \alpha_t}}{\sqrt{\alpha_t}}\epsilon.$$

Substituindo na função de perda ($N = 1$), temos que

$$\mathcal{L}(\theta)$$

$$\begin{aligned} &= \sum_{i=1}^I \left(\frac{1}{2\sigma_1^2} \left\| X^{(i)} - f_1(Z_1^{(i)}, \theta_1) \right\|^2 \right. \\ &\quad \left. + \sum_{t=2}^T \frac{1}{2\sigma_t^2} \left\| \left(\frac{1}{\sqrt{1 - \beta_t}} Z_t^{(i)} - \frac{\sqrt{\beta_t}}{\sqrt{1 - \alpha_t}\sqrt{1 - \beta_t}} \epsilon \right) - f_t(Z_t^{(i)}, \theta_t) \right\|^2 \right) \end{aligned}$$

Reparametrização da Loss Function II

Podemos mudar o modelo para prever o **ruído** ϵ

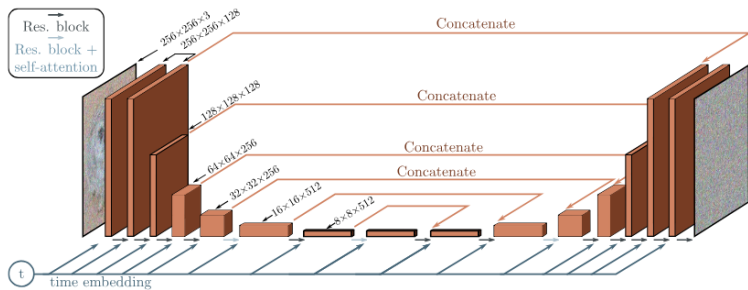
$$f_t[Z_t, \theta_t] = \frac{1}{\sqrt{1 - \beta_t}} Z_t - \frac{\sqrt{\beta_t}}{\sqrt{1 - \alpha_t} \sqrt{1 - \beta_t}} g_t(Z_t, \theta_t).$$

Substituindo na função de perda, temos que

$$\begin{aligned} \mathcal{L}(\theta) &= \sum_{i=1}^I \sum_{t=1}^T \left(\frac{\beta_t^2}{(1 - \alpha_t)(1 - \beta_t)2\sigma_t^2} \left\| g_t(Z_t^{(i)}, \theta_t) - \epsilon_t^{(i)} \right\|^2 \right) \\ &= \sum_{i=1}^I \sum_{t=1}^T \left(\frac{\beta_t^2}{(1 - \alpha_t)(1 - \beta_t)2\sigma_t^2} \left\| g_t(\sqrt{\alpha_t} X^{(i)} + \sqrt{1 - \alpha_t} \epsilon_t, \theta_t) - \epsilon_t \right\|^2 \right) \end{aligned}$$

Implementação

Podemos implementar g_t com uma **única** Unet usando time embeddings para t .



Implementação: Treino

- ▶ Repetir até a convergência:
 - ▶ Para cada $i \in \mathcal{B}$:
 - ▶ Amostra $t \sim \mathcal{U}(1, T)$
 - ▶ Amostra $\epsilon_t \sim \mathcal{N}(0, I)$
 - ▶ Calcula $\ell_i = \left\| g_t(\sqrt{\alpha_t} X^{(i)} + \sqrt{1 - \alpha_t} \epsilon_t, \theta_t) - \epsilon_t \right\|^2$
 - ▶ Calcula a média $\ell = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \ell_i$
 - ▶ Passo de descida do gradiente estocástico: $\theta \leftarrow \theta - \eta \nabla_{\theta} \ell$

Implementação: Geração de Imagens

- ▶ Amostra $Z_T \sim \mathcal{N}(0, I)$
- ▶ Para $t = T \dots, 1$:
 - ▶ Calcula $\hat{Z}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} Z_t - \frac{\sqrt{\beta_t}}{\sqrt{1-\alpha_t}\sqrt{1-\beta_t}} g_t(Z_t, \theta_t)$
 - ▶ Amostra $\epsilon \sim \mathcal{N}(0, I)$ se $t > 1$, senão $\epsilon = 0$
 - ▶ Calcula $Z_{t-1} = \hat{Z}_{t-1} + \sigma_t \epsilon$
- ▶ Retorna Z_0

Referências I

- [1] Simon Prince. *Understanding Deep Learning*. 2023. URL: <https://udlbook.github.io/udlbook/> (visited on 12/20/2023).